A counterexample to the unrestricted Sawyer-type conjecture

Automated mathematical discovery run `fa_banach_001`

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Status. Candidate counterexample, likely valid, subject to human review. The example disproves the version of Conjecture 5.11 allowing arbitrary $\varsigma > 0$. It does not address the alternative restriction $0 < \varsigma \le 1$.

1 Source question

The source is E. Roure-Perdices [1]. The paper is titled *Extrapolation via Sawyer-type inequalities*. Its arXiv identifier is 2404.09351v1, and the question is Conjecture 5.11 on pp. 31–32. The complete source statement is reproduced below as two full-width crops because it crosses a page boundary.

2]).

Theorem 5.10. *Fix $0 \le \alpha < \infty$, and let ν be a positive, measurable function. Given measurable functions f and g, suppose that for some exponents $1 < p < \infty$ and $\sigma > 0$, and every function $v \in \widehat{A}_p^\sigma$,*

$$\|g\|_{L^{p_\alpha,\infty}(V)} \le \psi(\|v\|_{\widehat{A}_p^\sigma})\|f\|_{L^{p,1}(v)},$$

where $\frac{1}{p_\alpha} = \frac{1}{p} + \alpha$, $V = v^{p_\alpha/p}\nu^{\alpha p_\alpha}$, and $\psi : [1,\infty) \longrightarrow [0,\infty)$ is an increasing function. Then, for every function $w \in A_1^\varsigma$ such that $W \in A_r^{\mathcal{R}}$ for some $r \ge 1$,

$$\|g\|_{L^{\frac{1}{1+\alpha},\infty}(W)} \le \Psi_r([w]_{A_1^\varsigma}, [W]_{A_r^{\mathcal{R}}})\|f\|_{L^{1,\frac{1}{p}}(w)},$$

where $\varsigma = \frac{\sigma p}{1+\sigma(p-1)}$, $W = w^{\frac{1}{1+\alpha}}\nu^{\frac{\alpha}{1+\alpha}}$, and $\Psi_r : [1,\infty)^2 \longrightarrow [0,\infty)$ is a function increasing in each variable.

It is worth mentioning that the following conjecture would allow us to prove Theorem 5.10 for an arbitrary exponent $1 \le q < p$.

Conjecture 5.11. *Fix exponents $q \ge 1$ and $\varsigma > 0$ (or $0 < \varsigma \le 1$), and write $\varrho = 1 + \varsigma(q-1)$. Let u and v be positive, measurable functions such that $u^\varsigma \in A_\varrho^{\mathcal{R}}$*

and $u^\varsigma v^\varrho \in A_r^{\mathcal{R}}$ for some $r \ge 1$. Then, there exists a function $\phi : [1,\infty)^2 \longrightarrow [0,\infty)$, increasing in each variable, such that for every measurable function f,

$$\left\|\frac{M(|f|^{q/\varrho}u^{\frac{1-\varsigma}{\varrho}})}{v}\right\|^{\varrho}_{L^{\varrho,\infty}(u^\varsigma v^\varrho)} \le \phi([u^\varsigma]_{A_\varrho^{\mathcal{R}}}, [u^\varsigma v^\varrho]_{A_r^{\mathcal{R}}})\|f\|^q_{L^{q,1}(u)}.$$

Remark 5.12. If $q = 1$ or $\varsigma = 1$, then Conjecture 5.11 follows from [74, Theorem 2] and [74, Lemma 3]. In general, for $h \in L^1_{loc}(\mathbb{R}^n)$ and $w \in A_\varrho^{\mathcal{R}}$, For clarity, write

$$\varrho = 1 + \varsigma(q-1).$$

The unrestricted form asserts that if $u^\varsigma \in A_\varrho^{\mathcal{R}}$ and $u^\varsigma v^\varrho \in A_r^{\mathcal{R}}$ for some $r \geq 1$, then

$$\left\| \frac{M\big(|f|^{q/\varrho} u^{(1-\varsigma)/\varrho}\big)}{v} \right\|_{L^{\varrho,\infty}(u^\varsigma v^\varrho)}^{\varrho} \leq \phi\Big([u^\varsigma]_{A_\varrho^{\mathcal{R}}}, [u^\varsigma v^\varrho]_{A_r^{\mathcal{R}}}\Big) \, \|f\|_{L^{q,1}(u)}^q, \tag{1}$$

with ϕ independent of f.

We use the source normalization

$$\|g\|_{L^{p,1}(w)} = p\int_0^\infty \lambda_g^w(t)^{1/p}\, dt, \qquad \|g\|_{L^{p,\infty}(w)} = \sup_{t>0} t\lambda_g^w(t)^{1/p}.$$

2 Proof intuition

At the sharp endpoint, the power weight $|x|^{p-1}$ is in the restricted class $A_p^{\mathcal{R}}$ even though it is not a classical A_p weight. This lets us make the maximal-function input exactly $x^{-1}\mathbf{1}_{[1,N]}$. Its average on $[0, N]$ is $(\log N)/N$, and the endpoint weight gives $[0, N]$ mass of order N^ϱ; hence the weak target quasi-norm to the ϱ-th power grows like $(\log N)^\varrho$. The source Lorentz norm of $x^{-1}\mathbf{1}_{[1,N]}$ grows only like $\log N$, so the right side of (1) grows like $(\log N)^q$. The exponent gap is $\varrho - q = (\varsigma - 1)(q - 1)$, which is positive precisely in the refuted range $\varsigma > 1$.

3 The counterexample

We first verify the only endpoint-weight fact needed in the construction.

Lemma 1. *For every $p > 1$, the weight $w_p(x) = |x|^{p-1}$ on $\mathbb{R}$ belongs to $A_p^{\mathcal{R}}$.*

Proof. We use the equivalent defining condition from the source:

$$\sup_I \sup_{E\subseteq I} \frac{|E|}{|I|}\left(\frac{w_p(I)}{w_p(E)}\right)^{1/p} < \infty, \tag{2}$$

where I ranges over intervals and E over measurable subsets of positive measure.

Put $L = |I|$ and $m = |E|$. Among all measurable subsets of $\mathbb{R}$ of measure m, the integral of the even increasing rearrangement $|x|^{p-1}$ is minimized by $[-m/2, m/2]$. Therefore

$$w_p(E) \geq \frac{m^p}{p, 2^{p-1}}. \tag{3}$$

If I meets the origin, then $|x| \leq L$ on I, so $w_p(I) \leq L^p$; (3) gives $w_p(E)/w_p(I) \geq (p2^{p-1})^{-1}(m/L)^p$.

Suppose next that $I = (a, b)$ lies in $(0,\infty)$; the negative case is symmetric. If $a < L$, then $b < 2L$, whence $w_p(I) \leq 2^{p-1}L^p$ and the preceding argument applies with another constant depending only on p. If $a \geq L$, then $b \leq 2a$ and

$$\frac{w_p(E)}{w_p(I)} \geq \frac{a^{p-1}m}{b^{p-1}L} \geq 2^{1-p}\frac{m}{L} \geq 2^{1-p}\left(\frac{m}{L}\right)^p.$$

Thus in every case $w_p(E)/w_p(I) \geq c_p(m/L)^p$, which is exactly the finiteness of (2). $\square$

Theorem 1. *For every $q > 1$ and $\varsigma > 1$, the estimate in the unrestricted $\varsigma > 0$ form of Conjecture 5.11 is false.*

Proof. Fix $q > 1$ and $\varsigma > 1$, and set

$$\varrho = 1 + \varsigma(q-1) > q, \qquad u(x) = |x|^{q-1}, \qquad v(x) = 1.$$

Then

$$u^{\varsigma}(x) = |x|^{\varsigma(q-1)} = |x|^{\varrho-1}.$$

Lemma 1, with $p = \varrho$, shows that $u^{\varsigma} \in A_{\varrho}^{\mathcal{R}}$. Since $v = 1$, the second hypothesis also holds by choosing $r = \varrho$. Both characteristics appearing in ϕ are fixed independently of the parameter below.

For $N > 2$, define

$$f_N(x) = x^{-1}\mathbf{1}_{[1,N]}(x).$$

The input to the maximal operator simplifies exactly:

$$\begin{aligned}|f_N(x)|^{q/\varrho} u(x)^{(1-\varsigma)/\varrho} &= x^{[-q+(q-1)(1-\varsigma)]/\varrho}\mathbf{1}_{[1,N]}(x)\\ &= x^{-1}\mathbf{1}_{[1,N]}(x),\end{aligned}$$

because $-q + (q-1)(1-\varsigma) = -\varrho$. For every $x \in [0, N]$, the interval $[0, N]$ is admissible in the uncentered Hardy–Littlewood maximal operator, and hence

$$M(x^{-1}\mathbf{1}_{[1,N]})(x) \geq \frac{\log N}{N}.$$

Taking a threshold half this value and using $u^{\varsigma}([0, N]) = N^{\varrho}/\varrho$ yields

$$\left\|M(x^{-1}\mathbf{1}_{[1,N]})\right\|_{L^{\varrho,\infty}(u^{\varsigma})}^{\varrho} \geq \frac{(\log N)^{\varrho}}{2^{\varrho}\varrho}. \tag{4}$$

It remains to bound the source norm. With respect to $u(x)dx$, the distribution function of f_N satisfies

$$\lambda_{f_N}^{u}(t) \leq \begin{cases} N^q/q, & 0 < t < N^{-1}, \\ (qt^q)^{-1}, & N^{-1} \leq t < 1, \\ 0, & t \geq 1. \end{cases}$$

Consequently

$$\|f_N\|_{L^{q,1}(u)} = q\int_0^{\infty} \lambda_{f_N}^{u}(t)^{1/q}\, dt \leq q^{1-1/q}(1 + \log N). \tag{5}$$

If (1) held, then (4) and (5), with the fixed weight characteristics absorbed into a constant, would imply

$$(\log N)^{\varrho} \leq C_{q,\varsigma}(1 + \log N)^q \qquad (N > 2).$$

This is impossible because $\varrho - q = (\varsigma - 1)(q - 1) > 0$. $\square$

4 Scope, stress tests, and verification

The construction disproves exactly the unrestricted clause $\varsigma > 0$. It does not disprove the parenthetical alternative $0 < \varsigma \leq 1$. Indeed the same calculation has target/source logarithmic exponent gap $(\varsigma - 1)(q - 1)$, so it is neutral at $\varsigma = 1$ (the case the source already records as true) and points in the non-obstructive direction for $\varsigma < 1$.

Three variants were checked. Adding logarithmic factors to the endpoint power weight changes both sides by the same slowly varying contribution and does not move the decisive gap. Replacing $v = 1$ by a power weight introduces extra compatibility constraints but does not extend this counterexample into $\varsigma \leq 1$. Finally, the second conjecture in Section 8.2 of the source has a different weighted-measure geometry and is not claimed to follow or fail from this example.

The separate `verification.md` audits each exponent and the direct $A_p^{\mathcal{R}}$ proof. The script `code/check_growth.py` prints finite-scale ratios for representative parameters; it is illustrative and is not used as proof.

5 Novelty audit and human-review recommendation

A bounded search on 13 August 2026 covered the exact paper title and arXiv id, "Conjecture 5.11", "Sawyer-type inequalities", "restricted weak type", and the notation $A_p^{\mathcal{R}}$, including the authors' earlier Sawyer-type paper [2]. No later arXiv paper proving or refuting this exact conjecture, and no occurrence of this endpoint power-weight counterexample, was located. The novelty claim is therefore provisional.

Human review is recommended. The two points deserving the closest check are Lemma 1 and the Lorentz normalization in (5). Both are elementary and fully reproduced, so verification does not depend on an external classification theorem.

References

[1] E. Roure-Perdices, *Extrapolation via Sawyer-type inequalities*, arXiv:2404.09351v1 (2024).

[2] C. Pérez and E. Roure-Perdices, *Sawyer-type inequalities for Lorentz spaces*, arXiv:2003.04167 (2020).